Scaling nested sampling for next generation gravitational-wave data

Metha Prathaban

About Me

- 3rd year PhD student in Will Handley's group
- Handley group works on Bayesian statistical inference and artificial intelligence methodologies, with a focus on analyzing complex datasets from next-generation surveys to explore a wide range of physics questions related to dark matter, dark energy, and the early Universe.
- My interests are all things Bayesian inference in the context of GWs!

The Handley group!

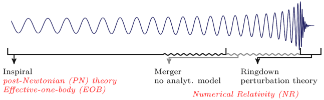
+ others!

Forward problem

model parameters
$\theta = \mathcal{M}, q, d_L, t_c \ldots$

waveform model

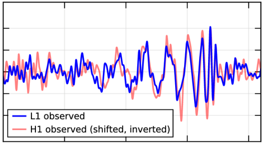
template for strain waveform



Inverse problem

observed signal strain

Livingston, Louisiana (L1)



inference

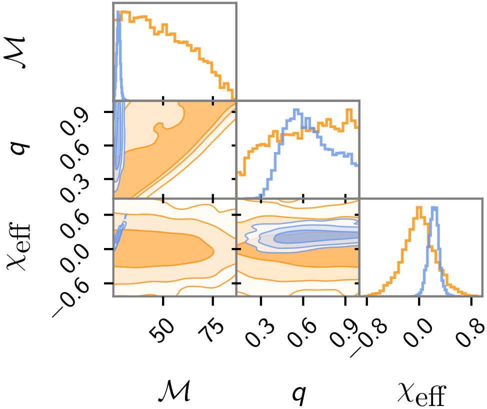
model parameters, inferred
θ_{est}

Given some model $\mathcal{M}$ and observed signal $\mathcal{D}$, Bayes' theorem enables us to relate the posterior probability of the set of parameters θ which generated the signal to the likelihood of the $\mathcal{D}$ given θ and the prior probability of θ given $\mathcal{M}$:

$$\mathcal{P}(\theta|D,\mathcal{M}) = \frac{P(D|\theta,\mathcal{M})P(\theta|\mathcal{M})}{P(D|\mathcal{M})} = \frac{\mathcal{L}(D|\theta)\pi(\theta)}{\mathcal{Z}} \tag{1}$$

The evidence, $\mathcal{Z}$, plays a key role in model comparison.

- Define prior, **sample** (unnormalized) posterior ($\mathcal{L}(\mathbf{d}|\theta) \times \pi(\theta)$).

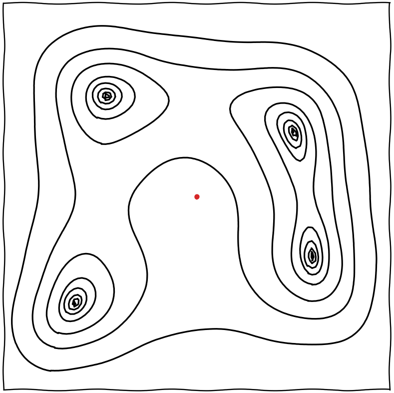


Challenges:

- High-dimensional parameter spaces $\Rightarrow$ posterior occupies vanishingly small region of prior.
- Complex waveform models with high costs

Need to sample posterior efficiently!

MCMC



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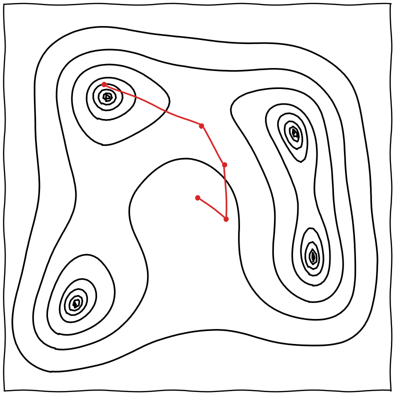

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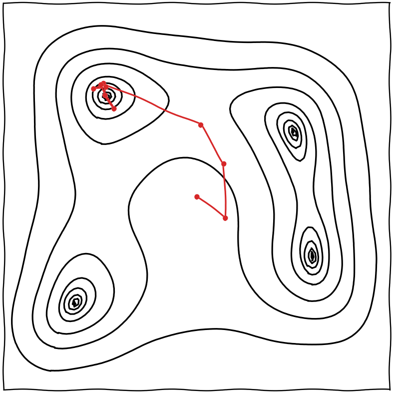

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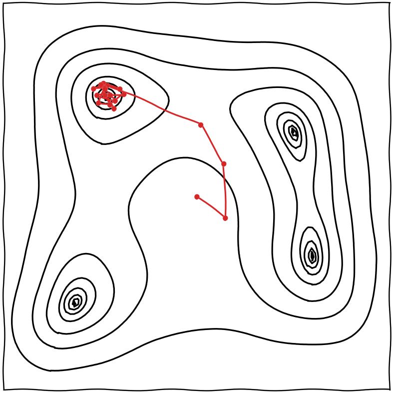

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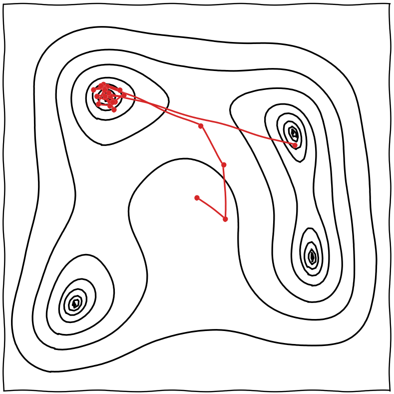

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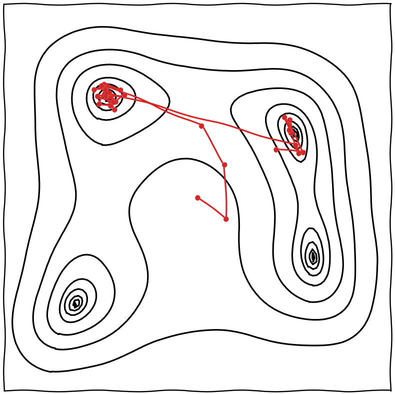

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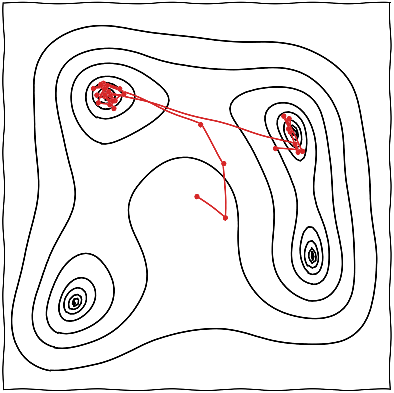
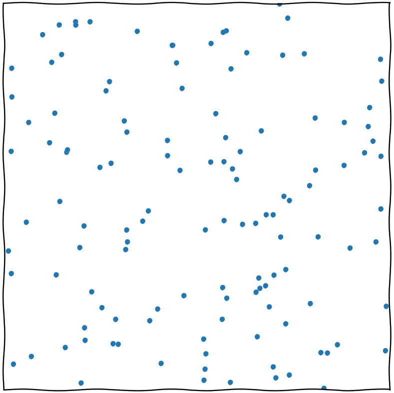


Nested sampling



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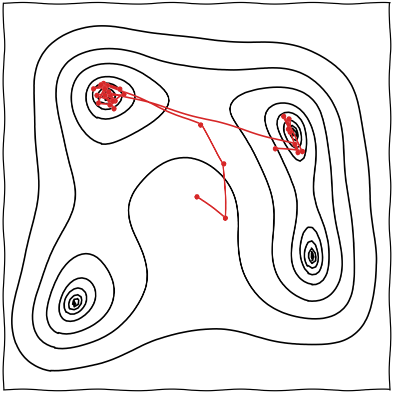

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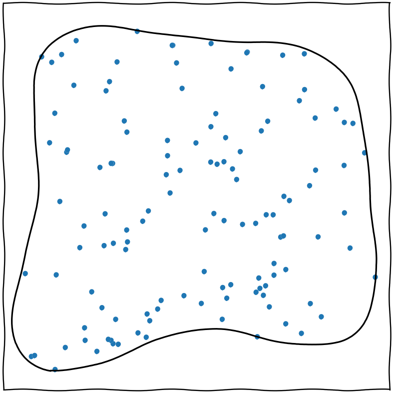

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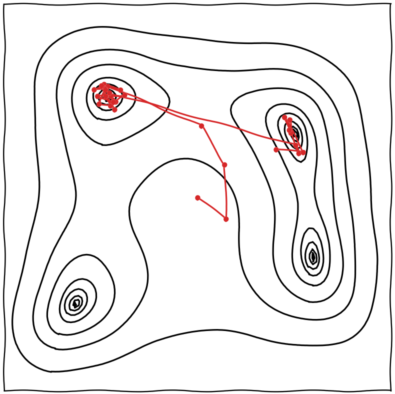

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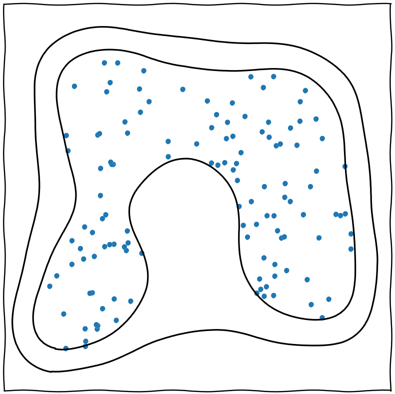

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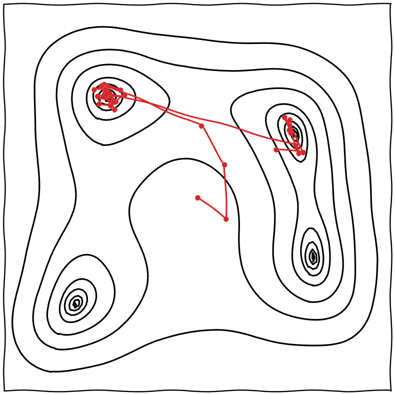

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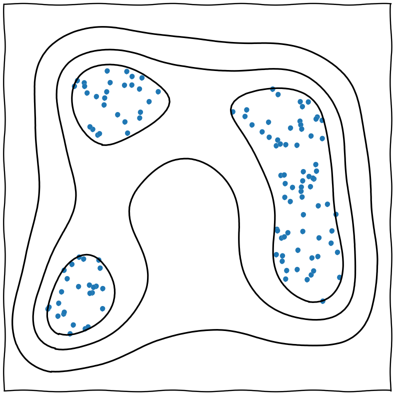

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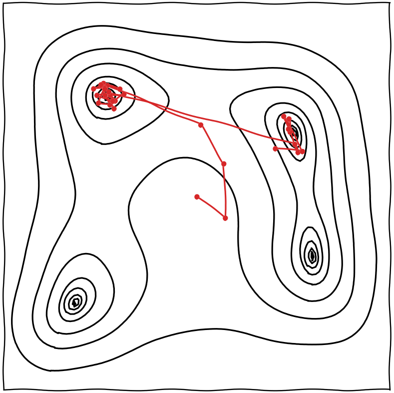

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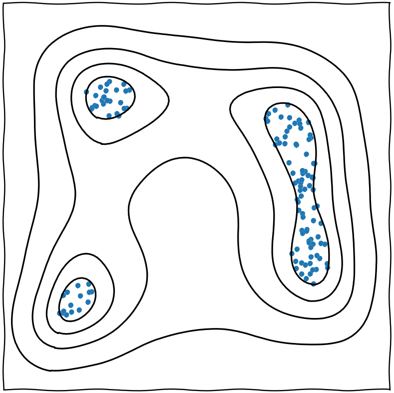

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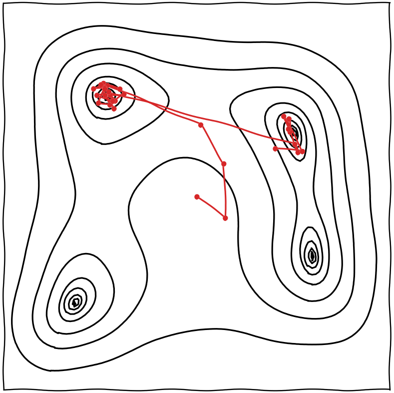

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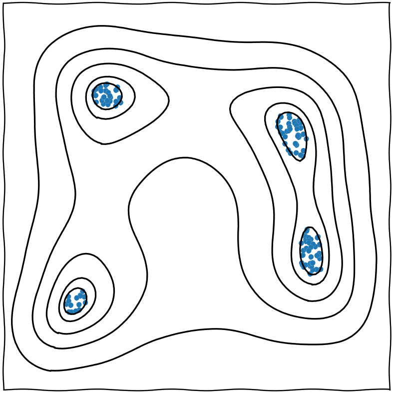

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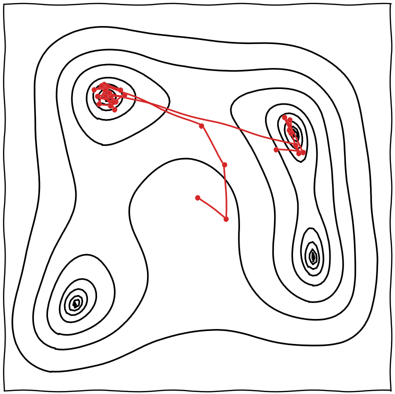

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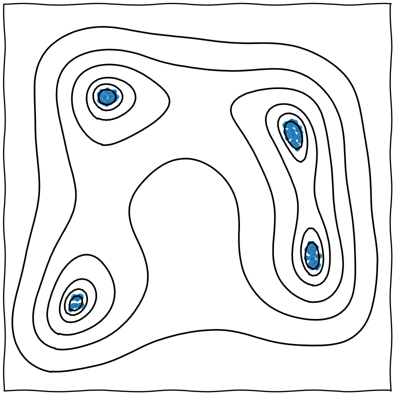

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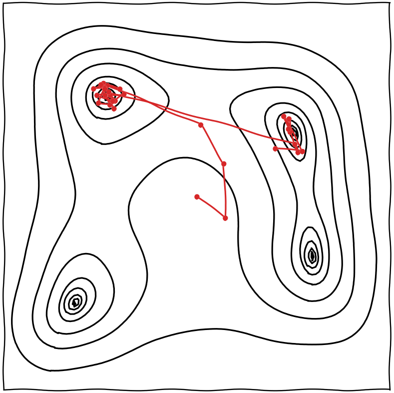

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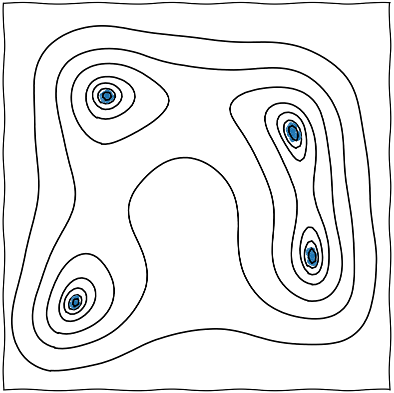

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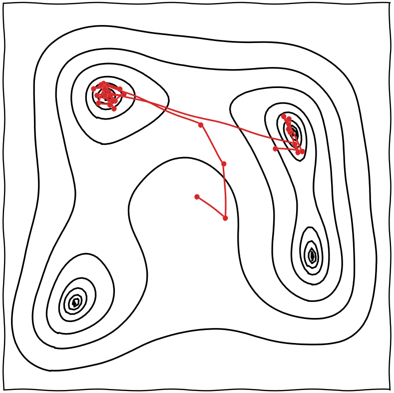

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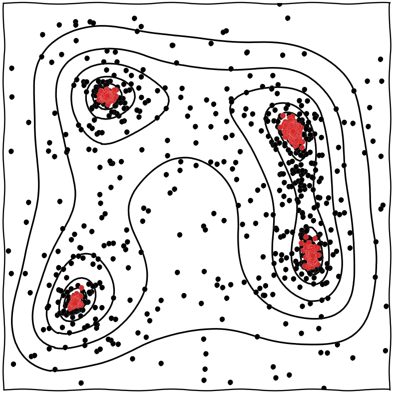
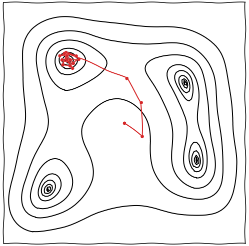


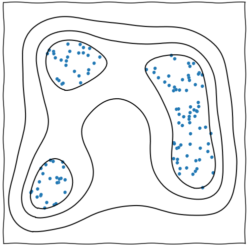
MCMC

- Single "walker", exploring posterior
- Fast, if proposal matrix is tuned
- Parameter estimation



Nested sampling

- Ensemble of "live points"
- Slower than MCMC methods
- Parameter estimation + model comparison



Current vs. future data

LVK data so far

- Short-duration signals from compact binary coalescences
- End of O4a: ≈ 200 confident detections
- Detection rate: ≈ 1 every few days
  - Signals not overlapping
- BBH mergers last seconds, BNS mergers last minutes
- Computationally intensive to analyse, but doable with current pipelines.

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ET + CE

- Ten-fold improvement in sensitivity compared to 2G detectors
- Detection rate: $\approx 10^5 - 10^6$ BBH and $\approx 7 \times 10^4$ BNS mergers per year
- Significantly higher SNR
- Longer signal durations
  - Overlapping signals expected
- New GW sources (not just CBCs)
- **Still doable with current pipelines?**

Time of convergence of NS: 2212.01760

likelihood evaluation time

$$T \propto T_{\mathcal{L}} \times f_{\text{sampler}} \times D_{\text{KL}} \times n_{\text{live}} \tag{2}$$

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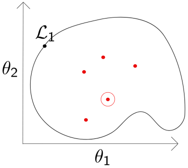

drawing new live point

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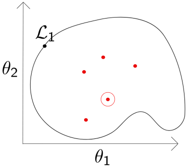
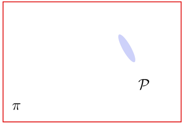
compression from prior to posterior $\left(\approx \ln \frac{V_\pi}{V_\mathcal{P}}\right)$

drawing new live point





Scaling of stochastic sampling

$$T_{\text{total}} \propto N_{\text{signals}} \times T_{\mathcal{L}} \times f_{\text{sampler}} \times D_{\text{KL}} \times n_{\text{live}} \quad (3)$$

higher sensitivity

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higher SNR + longer signal ⇒ narrower posterior

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longer signal duration $\Rightarrow$ slower waveform generation

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duration $\gtrsim 10\text{mins}$ $\Rightarrow$ extra modeling

Scaling of stochastic sampling

non-stationary noise

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Billions to quadrillions of CPU core hours for 1 month of ET data... 2412.02651


ROM, surrogates

$$T \propto T_{\mathcal{L}} \times f_{\text{sampler}} \times D_{\text{KL}} \times n_{\text{live}} \tag{4}$$

Machine learning accelerated NS

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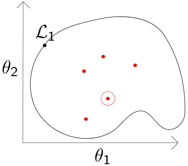
NESSAI



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not as baked-in as you might think!

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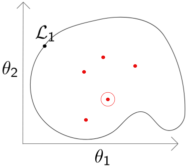

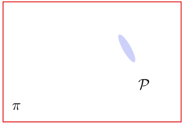

 

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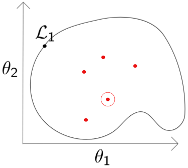
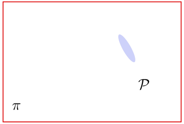




- **CPUs (Central Processing Units):**
  - Equipped with a limited number of powerful cores.
  - Optimised for wide range of tasks and excel at sequential, complex calculations and intricate decision-making.
- **GPUs (Graphics Processing Units):**
  - Architectured with thousands of smaller, efficient cores.
  - Best suited for repetitive, data-parallel tasks.

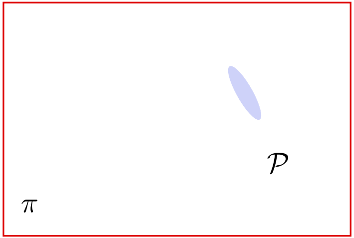
Advantages of GPUs for GW inference

- **Likelihood Evaluation:** Frequency bins are independent, so waveforms can be evaluated across many bins in parallel.
- **Sampling:** Multiple chains can be executed in parallel.

Conclusions

- 3G era presents significant challenges for data analysis
  - Current standard stochastic sampling methods, including traditional nested sampling, will not scale well
- ML-accelerated sampling and GPU-accelerated pipelines work towards addressing these challenges, and offer a complementary approach to SBI
  - GPUs are becoming the norm in many areas, including astrophysics, and computing power will become heavily skewed towards GPUs.
  - Adapting our algorithms to modern hardware is essential to keep pace with the cutting edge of computing power.

One way to do this (REACH):



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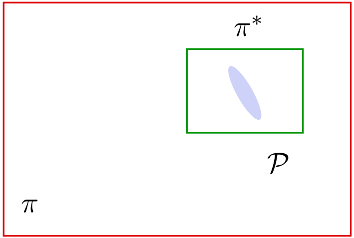

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Reducing D_{KL}

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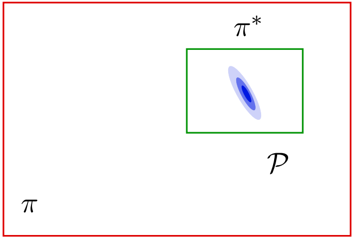

- Perform low resolution (low live points) run first to roughly identify where posterior lies.
- Then set off second, high resolution, run with **narrower** box prior (much quicker).
- Evidence has **changed** (since different prior), but easy to correct (multiply new evidence by $\frac{V_{\pi^*}}{V_\pi}$)

- Can iterate on this by using **normalizing flows** (NF) to learn the rough posterior.

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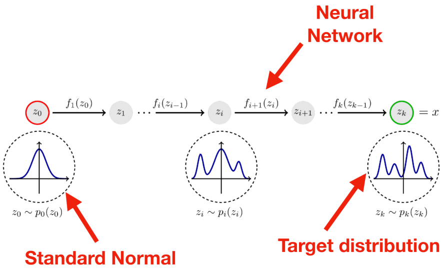
- NFs perform density estimation, by learning a series of invertible mappings from the standard normal distribution to the target (posterior).



- Use **normalizing flows** (NF) to learn the rough posterior, and use this as our updated prior, π^*.
- In this case, can't do our trick of correcting the second evidence by volume ratio, $\frac{V_{\pi^*}}{V_\pi}$!
- Must rely on another technique to get around this!

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Posterior repartitioning (PR) can help us with this! (see e.g. 2212.01760)

Bayesian Analysis (0000) **00**, Number 0, pp. 1

Bayesian posterior repartitioning for nested sampling

Xi Chen*,†, Farhan Feroz† and Michael Hobson†

Improving the efficiency and robustness of nested sampling using posterior repartitioning

Xi Chen · Michael Hobson · Saptarshi Das · Paul Gelderblom

Article

SuperNest: accelerated nested sampling applied to astrophysics and cosmology†

Aleksandr Petrosyan[1,2,3*] & Will Handley[1,2,4†]

- Evidence and posterior only depend on product of $\mathcal{L}$ and π:

$$\mathcal{Z} = \int \mathcal{L}(\theta)\pi(\theta)d\theta \qquad (5)$$

$$\mathcal{P}(\theta) = \frac{\mathcal{L}(\theta)\pi(\theta)}{\mathcal{Z}} \qquad (6)$$

We are free to redefine the likelihood and prior however we like - as long as the product is the same! arXiv:1908.04655

$$\tilde{\mathcal{Z}} = \int \tilde{\mathcal{L}}(\theta)\tilde{\pi}(\theta)d\theta = \int \mathcal{L}(\theta)\pi(\theta)d\theta = \mathcal{Z} \qquad (7)$$

PR (cont.)

- Many sampling algorithms do not distinguish between $\mathcal{L}$ and π at the algorithmic level.
- e.g. Metropolis-Hastings acceptance ratio only depends on the **joint distribution**, $\mathcal{L}(\theta)\pi(\theta)$.
- Nested sampling does distinguish between prior and likelihood at the algorithmic level, by 'sampling from the prior π, subject to the hard likelihood constraint, $\mathcal{L}$'.

- $\mathcal{Z}$ and $\mathcal{P}$ will not change if we repartition $\mathcal{L}$ and π, **but $\mathcal{D}_{\mathrm{KL}}$ will**.

$$\pi(\theta) \longrightarrow \mathsf{NF}(\theta)$$

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$$\mathcal{L}(\theta) \longrightarrow \frac{\mathcal{L}(\theta)\pi(\theta)}{\mathsf{NF}(\theta)}$$

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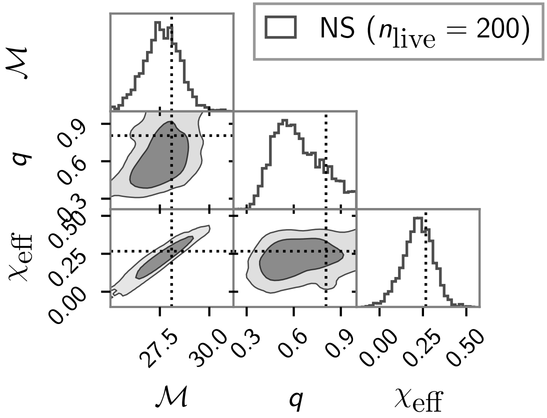
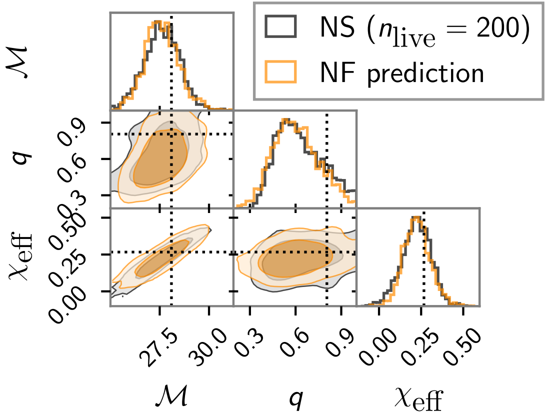
$$\mathcal{D}_{\mathrm{KL}} \approx \log \frac{V_{\mathsf{NF}}}{V_{\mathcal{P}}}$$

- Perform low resolution run on simulated data.

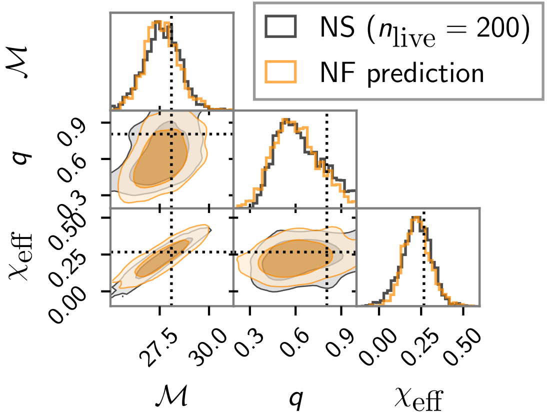


- Perform low resolution run on simulated data.
- Train NF on the weighted samples.

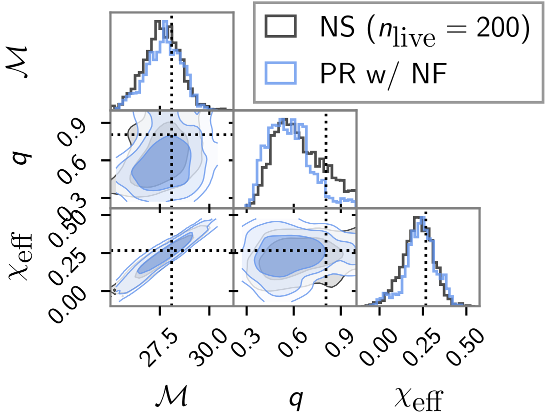


Demo on simulated example

- Perform low resolution run on simulated data.
- Train NF on the weighted samples.
- Use this as 'repartitioned prior' for high resolution run (and correct evidences and posteriors accordingly).



- Perform low resolution run on simulated data.
- Train NF on the weighted samples.
- Use this as 'repartitioned prior' for high resolution run (and correct evidences and posteriors accordingly).



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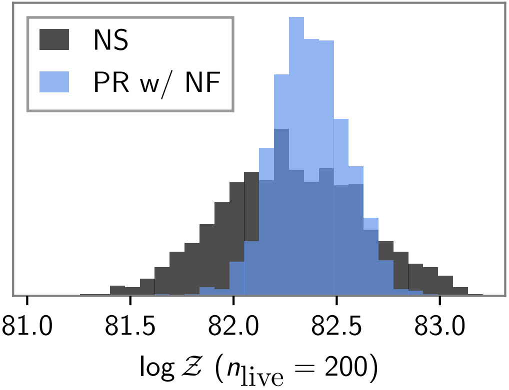
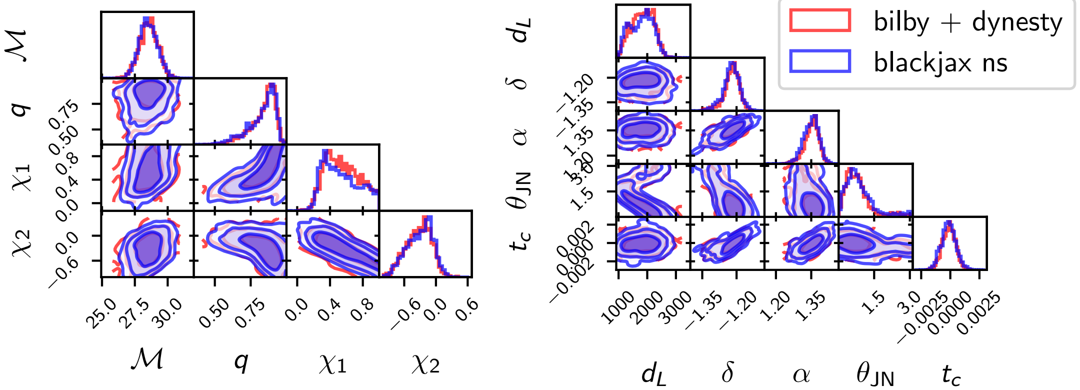


Same answer as doing a full resolution pass of NS, but 7x faster (precision-normalized).

BLACKJAX NS

- Nested sampling implementation in BLACKJAX, developed by David Yallup.
- Exploits parallel capabilites of GPUs to accelerate sampling procedure
- Combined with RIPPLE, can do GW inference on GPU
- Fast - 11 parameter nested sampling run on GW150914 in minutes
- Can run on CPUs too - many new features incorporated which enhance core nested sampling algorithm, and especially useful in context of GWs.



bilby + dynesty
blackjax ns
$\mathcal{M}$
q
χ_1
χ_2
d_L
δ
α
θ_{JN}
t_c